

RWS — Software Logic, Operation Sequence & Fire Control Document

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Change from Rev A: Added Section 12A (Mechanical Movement Verification Sensor), updated post-fire sequence, updated signal table, updated flow diagram, added Section 18 (Full System Hardware Architecture).

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1. Purpose & Scope

This document describes in full detail:

- The **software operational logic** of the RWS system
- The **sequence of states** from system power-on to weapon fire

- The **master/slave control relationship** between the electro-optical camera gimbal and the pan-tilt weapon platform
- The **fire control interlock sequence** and all conditions that must be satisfied before solenoid actuation
- The **operator display symbology** during all phases

This document is intended for use by the software development team, system integrator, safety engineer, and verification engineer.

2. System Context & Key Architectural Assumptions

The following architectural facts are foundational to all logic described in this document and must be understood before reading any further.

2.1 Two Separate Platforms — Critical Distinction

The RWS is installed on a **drone platform**. The drone has its own independent flight control system, its own flight camera, and its own operator control — all of which are **entirely separate** from the RWS. The RWS has **no interface with, no dependency on, and no knowledge of** the drone's flight software, attitude controller, or flight camera. The RWS operates as a **fully autonomous payload** from the drone's perspective.

2.2 Two Separate Camera Systems

System	Purpose	Controlled by
Drone flight camera	Drone navigation and piloting	Drone operator (separate GCS)
RWS EO/IR camera (on elevation platform)	Target observation and weapon slaving	RWS operator (RWS GCS, Joystick 1)

These two camera systems are **never mixed, never shared, and never co-commanded**.

2.3 Camera Gimbal is a Self-Contained Stabilized Unit

The EO/IR camera assembly is mounted on the **elevation platform** of the RWS pan-tilt system. It has its **own independent gimbal motors and its own internal stabilization system**. This camera gimbal:

- Stabilizes itself against drone vibration and motion using its own internal IMU
- Receives slew commands from the RWS operator via Joystick 1
- Reports its absolute pointing direction (azimuth and elevation angles) to the Jetson FCC at all times
- Acts as the **master** in the master/slave fire control loop

2.4 Pan-Tilt is the Slave

The RWS pan-tilt platform (AZ axis + EL axis) is the **slave** in the fire control loop:

- In normal observation, it remains **stationary at a fixed forward-horizon position**
- It only moves when commanded to slave to the camera's locked coordinates
- It receives its move commands from the Jetson FCC (not directly from the operator joystick)

2.5 The Jetson FCC is the Brain

The Jetson A (FCC Primary) is the central processing node that:

- Receives camera pointing data continuously
- Receives pan-tilt sensor feedback continuously (AZ heading sensor + EL angle sensor)
- Computes the angular error between camera aim and pan-tilt pointing
- Sends correction commands to the Gimbal Controller → AZ servo + EL servo
- Manages the fire control state machine
- Drives all operator display symbology

3. Subsystem Roles & Responsibilities

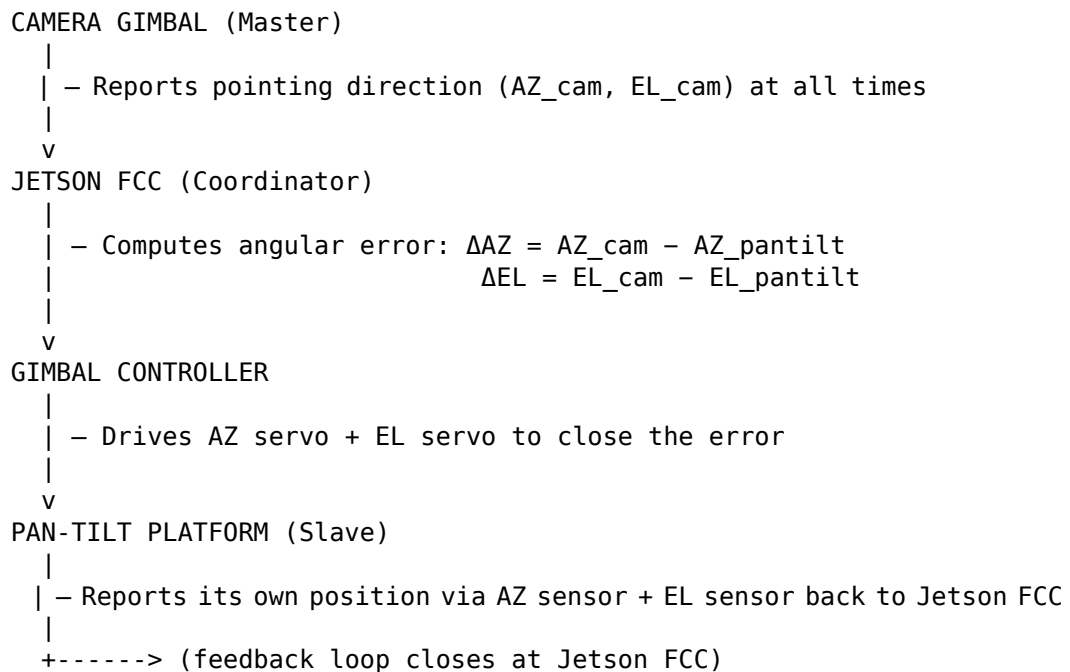
Subsystem	Role in System	Inputs	Outputs
RWS EO/IR Camera Gimbal	Master — observation and target lock	Joystick 1 commands, internal IMU	Camera AZ/EL coordinates to Jetson FCC

Subsystem	Role in System	Inputs	Outputs
Jetson A (FCC Primary)	Central logic, state machine, coordinate processing	Camera coordinates, AZ sensor, EL sensor, operator buttons	Commands to Gimbal Controller, display symbology, ISM authorization
Gimbal Controller	Pan-tilt motor servo controller	Jetson FCC AZ/EL commands, AZ/EL sensor feedback	AZ servo motor command, EL servo motor command
AZ Servo	Azimuth axis actuator	Gimbal Controller CAN command	Azimuth rotation of pan-tilt platform
EL Servo (BLDC, 1:60 worm gear)	Elevation axis actuator	Gimbal Controller CAN command	Elevation rotation of pan-tilt platform
Azimuth Sensor	AZ absolute position feedback	Physical azimuth position	AZ heading angle → Jetson FCC
EL Angle Sensor	EL absolute position feedback	Physical elevation position	EL angle → Jetson FCC
EL Direction Sensor	EL incremental direction	Physical elevation motion	EL direction → Jetson FCC
ISM Safety MCU	Independent safety authority	Jetson FCC authorization signal (discrete), sensors	Solenoid actuation signals
Safety Open Solenoid	Releases weapon safety	ISM command	Mechanical safety open
Safety Close Solenoid	Engages weapon safety	ISM command / failsafe	Mechanical safety close
Safety Pin Solenoid	Master trigger enable	ISM command	Pin actuation
Trigger Solenoid	Weapon fire	ISM command	Trigger actuation

Subsystem	Role in System	Inputs	Outputs
Operator GCS (10")	Human interface	Joystick 1, Button 1, Trigger button	Display output to operator

4. Master / Slave Relationship — Camera vs Pan-Tilt

4.1 Concept



4.2 Coordinate Convention

Axis	Zero Reference	Positive Direction	Sensor
Azimuth (AZ)	Forward (nose of drone / front of mount)	Clockwise when viewed from above	Azimuth sensor
Elevation (EL)	Horizontal (0°)	Upward (positive elevation)	EL angle sensor

All angles are expressed in degrees with one decimal place resolution minimum.

4.3 Coordinate Data Flow

Camera Gimbal → [AZ_cam, EL_cam] → Jetson FCC (100 Hz update rate minimum)
AZ Sensor → [AZ_pantilt] → Jetson FCC (100 Hz update rate minimum)
EL Sensor → [EL_pantilt] → Jetson FCC (100 Hz update rate minimum)

Jetson FCC computes:

$$\Delta AZ = AZ_cam - AZ_pantilt$$

$$\Delta EL = EL_cam - EL_pantilt$$

Alignment condition (boresight achieved):

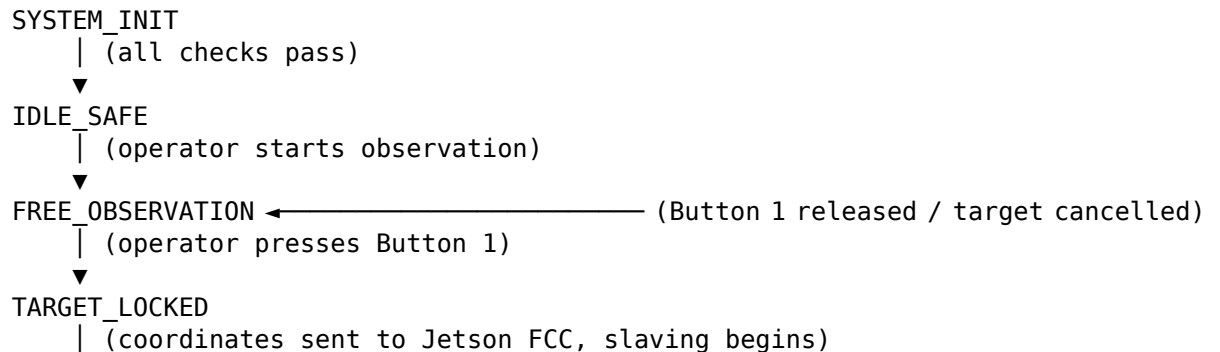
$$|\Delta AZ| \leq 0.5^\circ \text{ AND } |\Delta EL| \leq 0.5^\circ \text{ (threshold TBD per weapon accuracy requirement)}$$

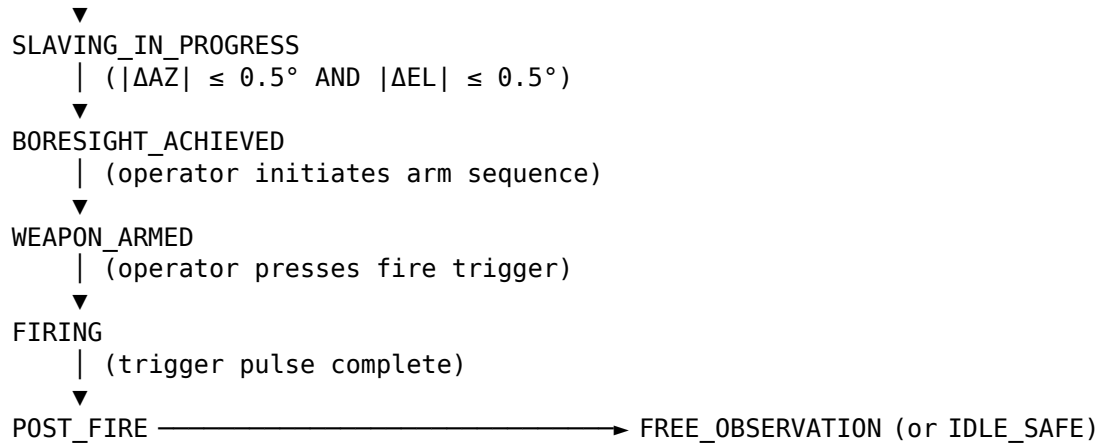
5. Software State Machine — Full System

The Jetson FCC runs a top-level state machine with the following states:

STATE 0 : SYSTEM_INIT
STATE 1 : IDLE_SAFE
STATE 2 : FREE_OBSERVATION
STATE 3 : TARGET_LOCKED
STATE 4 : SLAVING_IN_PROGRESS
STATE 5 : BORESIGHT_ACHIEVED
STATE 6 : WEAPON_ARMED
STATE 7 : FIRING
STATE 8 : POST_FIRE
STATE F : FAULT / ABORT

State transition overview:





Any state → FAULT/ABORT (on: link loss, sensor fault, overheat, recoil anomaly, operator fault)
 FAULT/ABORT → IDLE_SAFE (after fault cleared and operator acknowledges)

6. Phase 1 — System Initialization & Idle

6.1 Power-On Sequence

Upon system power-on, the following initialization sequence executes automatically:

- STEP 1 Power rails stabilize (24V main bus, compute rail, servo rail, safety rail)
- STEP 2 ISM Safety MCU boots – runs self-test independently
 - Confirms all solenoids are de-energized (SAFE state)
 - Confirms weapon temperature within safe range
 - Confirms recoil sensor nominal
- STEP 3 Jetson A boots (≤ 10 seconds to operational)
- STEP 4 Jetson B boots as hot standby
- STEP 5 Gimbal Controller initializes – queries AZ sensor + EL sensor
- STEP 6 EL servo and AZ servo perform homing / reference check
- STEP 7 Camera gimbal initializes its internal stabilization loop
- STEP 8 RF link established with GCS – handshake confirmed
- STEP 9 Optic Master initializes EO camera and IR thermal camera streams
- STEP 10 Jetson A sends "SYSTEM READY" signal to GCS

6.2 Pan-Tilt Default Position

Upon completion of initialization, the pan-tilt platform **moves to and locks at the fixed forward-horizon position:**

AZ_default = 0.0° (forward)

EL_default = 0.0° (horizon)

The pan-tilt holds this position rigidly using the worm gear's self-locking property. The EL and AZ servo drives are powered but commanded to hold position. The platform does **not respond to Joystick 1** — the joystick controls only the camera gimbal at this stage.

6.3 Entry Conditions to IDLE_SAFE

- All solenoids confirmed de-energized
 - ISM in SAFE state
 - Pan-tilt at forward-horizon default
 - Camera streams active on GCS display
 - RF link confirmed with latency ≤ 50 ms
-

7. Phase 2 — Free Observation Mode (Joystick 1 / Camera Roaming)

7.1 Description

In FREE_OBSERVATION state, the operator uses **Joystick 1** to freely slew the EO/IR camera gimbal in all directions — azimuth and elevation — to observe the area of interest. The camera moves independently on its own stabilized gimbal.

7.2 Pan-Tilt Behavior

- The pan-tilt platform **remains stationary at the forward-horizon default position (AZ=0°, EL=0°)**
- The pan-tilt does not track the camera in this phase
- The pan-tilt motors are powered and holding position via servo lock

7.3 Camera Behavior

- Camera gimbal receives Joystick 1 slew commands
- Camera gimbal stabilizes its own pointing against drone vibration and motion
- Camera reports its current pointing angles (AZ_cam, EL_cam) to Jetson FCC at ≥ 100 Hz
- Operator views live EO or IR video feed on GCS 10" display
- The operator can switch between EO (day) and IR (thermal) feeds at any time

7.4 Jetson FCC Behavior

- Continuously receives and logs (AZ_cam, EL_cam) from camera
- Continuously receives and logs (AZ_pantilt, EL_pantilt) from AZ and EL sensors
- Does **not** issue any slaving commands in this state
- Monitors all system health parameters
- Monitors RF link quality

7.5 Operator Display — FREE_OBSERVATION

```
LIVE VIDEO FEED (E0 or IR)

Crosshair at camera center (white)

AZ_cam: 034.5°    EL_cam: -12.3°
PAN-TILT: HOLD    FORWARD/HORIZON

STATE: FREE OBSERVATION
WEAPON: SAFE
```

8. Phase 3 — Target Lock (Button 1)

8.1 Trigger Condition

The operator identifies a target of interest via the camera feed and **presses Button 1**.

8.2 Camera Behavior on Button 1

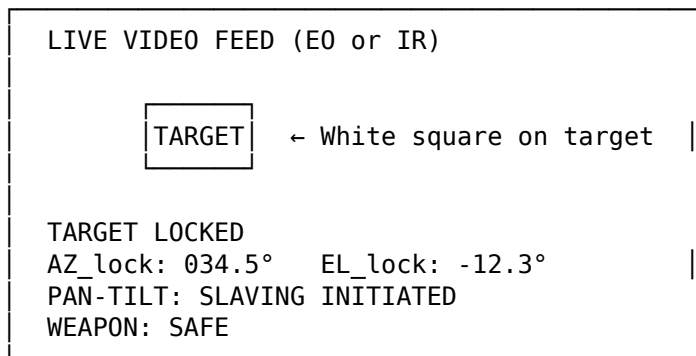
Upon Button 1 press, the camera gimbal executes the following immediately:

- ```
STEP 1 Camera freezes its current pointing direction
 AZ_lock = AZ_cam at moment of button press
 EL_lock = EL_cam at moment of button press
STEP 2 Camera activates internal target lock / position hold mode
 → Camera stabilizes on (AZ_lock, EL_lock) –
does NOT follow any further joystick input
 → Camera internal stabilization loop now holds this absolute pointing direction
STEP 3 Camera reports confirmed lock coordinates to Jetson FCC:
 MESSAGE: TARGET_LOCK_CONFIRMED { AZ_lock, EL_lock, timestamp }
```

### 8.3 Jetson FCC Behavior on TARGET\_LOCK\_CONFIRMED

STEP 1 Jetson FCC receives and stores (AZ\_lock, EL\_lock)  
STEP 2 Jetson FCC transitions state machine to STATE 3: TARGET\_LOCKED  
STEP 3 Jetson FCC sends locked coordinates to Gimbal Controller as the slaving target  
STEP 4 Jetson FCC sends display update to GCS:  
→ Draw a WHITE SQUARE around the locked target position on the video feed  
→ Display locked coordinates numerically on the HUD

### 8.4 Operator Display — TARGET\_LOCKED



## 9. Phase 4 — Pan-Tilt Slaving to Camera Coordinates

### 9.1 Description

Immediately after TARGET\_LOCK\_CONFIRMED, the Jetson FCC commands the Gimbal Controller to move the pan-tilt platform from its current position (forward-horizon) toward the locked camera coordinates (AZ\_lock, EL\_lock). The pan-tilt now acts as the slave, closing the angular error gap.

During this slaving phase the IMU is **not yet active** as a stabilization source — the pan-tilt is in gross motion and IMU stabilization only activates once boresight is achieved (see Phase 5).

### 9.2 Slaving Control Logic (Jetson FCC)

LOOP (running at  $\geq 100$  Hz):

READ AZ\_pantilt from azimuth sensor  
READ EL\_pantilt from EL angle sensor

```

COMPUTE:
 ΔAZ = AZ_lock - AZ_pantilt
 ΔEL = EL_lock - EL_pantilt

IF |ΔAZ| > 0.5° OR |ΔEL| > 0.5°:
 SEND to Gimbal Controller:
 AZ_command = AZ_lock (absolute position target)
 EL_command = EL_lock (absolute position target)
 IMU_STABILIZATION = OFF
 STATE = SLAVING_IN_PROGRESS

IF |ΔAZ| ≤ 0.5° AND |ΔEL| ≤ 0.5°:
 SEND to Gimbal Controller: HOLD_POSITION
 STATE = BORESIGHT_ACHIEVED (go to Phase 5)

END LOOP

```

### 9.3 Gimbal Controller Behavior

The Gimbal Controller executes the position servo loop:

```

FOR AZ axis:
 Error_AZ = AZ_command - AZ_sensor_feedback
 AZ_motor_velocity = Kp_AZ × Error_AZ + Kd_AZ × d(Error_AZ)/dt
 Send CAN velocity command to AZ servo drive

FOR EL axis:
 Error_EL = EL_command - EL_sensor_feedback
 EL_motor_velocity = Kp_EL × Error_EL + Kd_EL × d(Error_EL)/dt
 Send CAN velocity command to EL servo drive (BLDC, 1:60 worm gear)

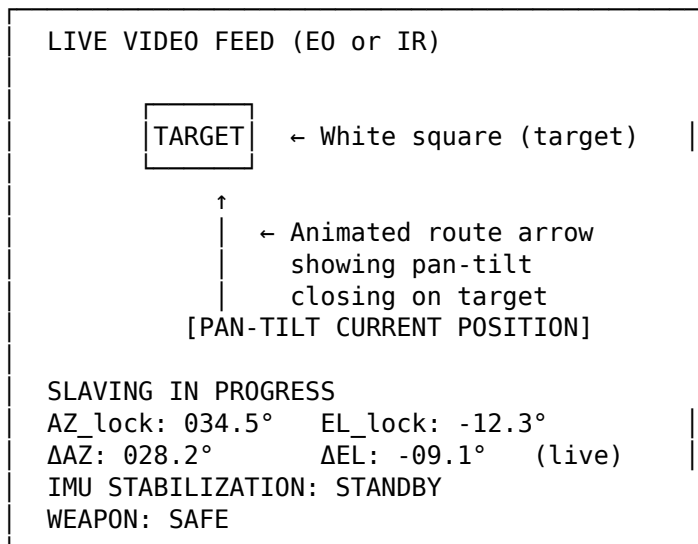
```

### 9.4 Pan-Tilt Motion Constraints

| Parameter             | Value                                                                      |
|-----------------------|----------------------------------------------------------------------------|
| Maximum AZ slew speed | ≥ 22.5 °/s (MIL-STD-1472H)                                                 |
| Maximum EL slew speed | Full arc ≤ 3 seconds<br>(MIL-STD-1472H)                                    |
| Motion profile        | Trapezoidal velocity profile<br>(controlled acceleration and deceleration) |
| Overshoot             | ≤ 0.5° on both axes                                                        |
| Approach behavior     | Decelerate as                                                              |

## 9.5 Operator Display — SLAVING\_IN\_PROGRESS

During pan-tilt slaving, the GCS display shows the operator a **live routing indicator**:



**Display elements during SLAVING\_IN\_PROGRESS:** - White square remains on locked target position - An animated arrow or progress indicator moves from the pan-tilt's current position toward the target square - Live  $\Delta AZ$  and  $\Delta EL$  values displayed numerically and decrease in real time - State label: "SLAVING IN PROGRESS" - IMU status: "STANDBY" (not yet active) - Weapon status remains: SAFE

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## 10. Phase 5 — Boresight Achieved / IMU Stabilization Active

### 10.1 Trigger Condition

$$|\Delta AZ| \leq 0.5^\circ \quad \text{AND} \quad |\Delta EL| \leq 0.5^\circ$$

Both conditions must be true **simultaneously and sustained for  $\geq 500$  ms** before boresight is declared confirmed. This prevents false confirmation due to mechanical overshoot or sensor noise.

## 10.2 IMU Stabilization Loop — Activation

This is the most critical change from Phase 4. The moment bore-sight is confirmed, the Jetson FCC **activates the IMU stabilization loop** for the pan-tilt AZ and EL servos. From this point forward, the IMU continuously measures the drone's angular rates and body motion, and the Jetson FCC uses these measurements to issue real-time correction commands to the AZ and EL servos, keeping the weapon pointed at (AZ\_lock, EL\_lock) in **world-fixed space** regardless of drone movement.

**The IMU stabilization loop operates as follows:**

IMU provides at  $\geq 1$  kHz:

$\omega_x$  = roll rate (rad/s)  
 $\omega_y$  = pitch rate (rad/s)  
 $\omega_z$  = yaw rate (rad/s)

Jetson FCC IMU Stabilization Loop (runs at  $\geq 500$  Hz):

READ IMU angular rates: [ $\omega_x$ ,  $\omega_y$ ,  $\omega_z$ ]

TRANSFORM drone angular rates to pan-tilt axis corrections:

$\delta AZ_{imu} = -\omega_z \times dt$  (yaw of drone disturbs AZ directly)  
 $\delta EL_{imu} = -(\omega_y \times \cos(AZ_{lock}) - \omega_x \times \sin(AZ_{lock})) \times dt$   
(pitch and roll of drone disturb EL depending on current AZ)

COMPUTE corrected servo target:

$AZ_{corrected} = AZ_{lock} + \Sigma(\delta AZ_{imu})$  (integrated correction)  
 $EL_{corrected} = EL_{lock} + \Sigma(\delta EL_{imu})$  (integrated correction)

SEND to Gimbal Controller:

$AZ_{command} = AZ_{corrected}$   
 $EL_{command} = EL_{corrected}$

VERIFY via sensors:

READ  $AZ_{pantilt}$ ,  $EL_{pantilt}$   
 $Residual_{AZ} = AZ_{corrected} - AZ_{pantilt}$   
 $Residual_{EL} = EL_{corrected} - EL_{pantilt}$   
→ Servo loop closes the residual at  $\geq 100$  Hz

IF  $|Residual_{AZ}| > 1.5^\circ$  OR  $|Residual_{EL}| > 1.5^\circ$ :

→ Drone disturbance exceeds servo correction capability  
→ Jetson FCC sends STABILIZATION\_WARNING to GCS  
→ IF sustained > 2 seconds: exit BORESIGHT\_ACHIEVED → re-slave

END LOOP

### 10.3 IMU Stabilization Loop — Key Properties

| Property                              | Value                                                         |
|---------------------------------------|---------------------------------------------------------------|
| IMU data rate                         | $\geq 1$ kHz                                                  |
| Stabilization correction rate         | $\geq 500$ Hz                                                 |
| Servo position correction rate        | $\geq 100$ Hz                                                 |
| Axes corrected                        | AZ (yaw) + EL (pitch/roll coupled)                            |
| Maximum correctable drone disturbance | $\pm 5^\circ/\text{s}$ angular rate (servo bandwidth limited) |
| Stabilization residual error          | $\leq 0.3^\circ$ RMS under normal drone hover disturbance     |
| IMU drift correction                  | Low-pass filter on integrated IMU bias to prevent drift       |

### 10.4 IMU Stabilization Loop — Deactivation Condition

The IMU stabilization loop is **active only when TARGET\_LOCK is engaged**. It is immediately deactivated under any of the following conditions:

| Condition                                                             | IMU Loop Action                                            |
|-----------------------------------------------------------------------|------------------------------------------------------------|
| Operator releases TARGET LOCK (Button 1 pressed again)                | STOP — IMU loop opens, pan-tilt returns to forward-horizon |
| System enters FAULT / ABORT state                                     | STOP — pan-tilt holds last position passively              |
| System returns to FREE_OBSERVATION state                              | STOP — IMU loop closed                                     |
| Drone angular rate exceeds $\pm 15^\circ/\text{s}$ (structural limit) | STOP — stabilization warning issued, re-slave required     |

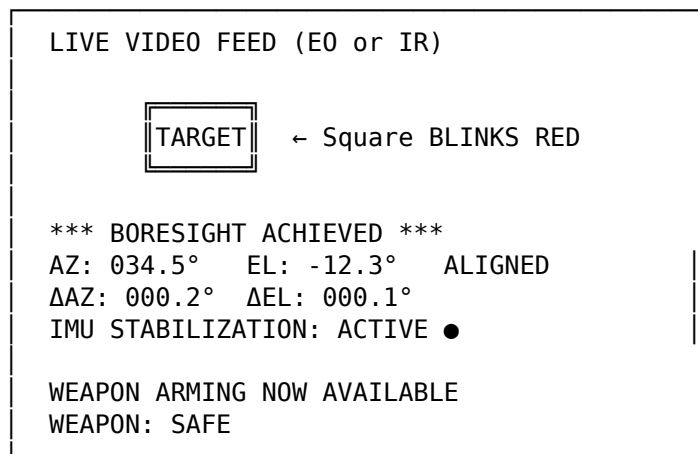
When the IMU loop stops, the pan-tilt **does not move abruptly** — the Gimbal Controller receives a HOLD\_LAST\_POSITION command and the worm gear passive self-lock maintains position until the next command.

### 10.5 Jetson FCC Actions on Boresight Achieved

STEP 1 Jetson FCC transitions to STATE 5: BORESIGHT\_ACHIEVED  
STEP 2 Jetson FCC activates IMU stabilization loop

- IMU corrections begin feeding into AZ and EL servo commands
- Worm gear provides passive mechanical hold between corrections
- STEP 3 Jetson FCC sends BORESIGHT\_ACHIEVED event to GCS display
  - "IMU STABILIZATION: ACTIVE" shown on HUD in green
- STEP 4 Jetson FCC sends BORESIGHT\_ACHIEVED to ISM Safety MCU (discrete signal)
  - ISM acknowledges but does NOT arm yet
  - ISM awaits further operator arm command

## 10.6 Operator Display — BORESIGHT\_ACHIEVED + IMU ACTIVE



**Key display changes:** - Target square changes from white static to **blinking red** — boresight confirmed - New HUD element: **"IMU STABILIZATION: ACTIVE"** in green with a pulsing green dot — this tells the operator the pan-tilt is actively compensating for drone motion - The operator can observe the pan-tilt micro-corrections happening in real time via the  $\Delta$ AZ/ $\Delta$ EL values remaining near zero despite drone movement

## 11. Phase 6 — Weapon Arming Sequence

### 11.1 Entry Condition

Weapon arming is **only permitted** when **ALL** of the following conditions are simultaneously true:

| Condition                  | Source                   | Required State |
|----------------------------|--------------------------|----------------|
| STATE = BORESIGHT_ACHIEVED | Jetson FCC state machine | True           |

| Condition                            | Source         | Required State   |
|--------------------------------------|----------------|------------------|
|                                      | $\Delta AZ$    | $\leq 0.5^\circ$ |
|                                      | $\Delta EL$    | $\leq 0.5^\circ$ |
| Weapon temperature $\leq T_{max}$    | ISM Safety MCU | True             |
| Recoil sensor nominal                | ISM Safety MCU | True             |
| RF link active, latency $\leq 50$ ms | Jetson FCC     | True             |
| ISM Safety MCU in SAFE state         | ISM            | True             |

If **any** of the above conditions is not met, the arm command from the operator is **rejected** and the display shows a fault reason.

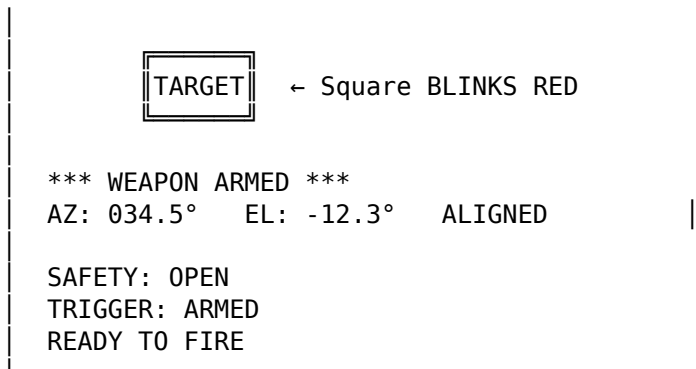
## 11.2 Arm Command Sequence

The operator initiates arming by pressing the **Arm Button** (dedicated hardware button, not touchscreen):

- STEP 1 Operator presses Arm Button
- STEP 2 Jetson FCC checks all entry conditions (table above)
- STEP 3 IF all conditions met:  
     Jetson FCC sends ARMING\_AUTHORIZATION (discrete signal) to ISM Safety MCU  
     ELSE:  
     Jetson FCC sends ARMING\_REJECTED + fault reason to GCS display  
     Remain in BORESIGHT\_ACHIEVED state
- STEP 4 ISM Safety MCU receives ARMING\_AUTHORIZATION  
     ISM independently verifies:  
     → Weapon temperature check  
     → Recoil sensor check  
     → All discrete safety inputs nominal
- STEP 5 IF ISM verification passes:  
     ISM energizes SAFETY OPEN SOLENOID → opens weapon mechanical safety  
     ISM energizes SAFETY PIN SOLENOID → arms master trigger  
     ISM transitions to ARMED state  
     ISM sends ARMED\_CONFIRMED (discrete signal) to Jetson FCC
- STEP 6 Jetson FCC receives ARMED\_CONFIRMED  
     Jetson FCC transitions to STATE 6: WEAPON\_ARMED  
     Jetson FCC sends ARMED display update to GCS

## 11.3 Operator Display — WEAPON\_ARMED

|                            |
|----------------------------|
| LIVE VIDEO FEED (EO or IR) |
|----------------------------|



## 12. Phase 7 — Fire Sequence

### 12.1 Entry Condition

Fire is **only permitted** when **ALL** of the following are simultaneously true:

| Condition                         | Source         | Required State                   |
|-----------------------------------|----------------|----------------------------------|
| STATE = WEAPON_ARMED              | Jetson FCC     | True                             |
| ISM in ARMED state                | ISM Safety MCU | True                             |
|                                   | $\Delta AZ$    | $\leq 0.5^\circ$ (still aligned) |
|                                   | $\Delta EL$    | $\leq 0.5^\circ$ (still aligned) |
| RF link active                    | Jetson FCC     | True                             |
| Weapon temperature $\leq T_{max}$ | ISM Safety MCU | True                             |

If alignment drifts beyond  $0.5^\circ$  after arming (e.g., due to drone disturbance), the system **automatically exits WEAPON\_ARMED** back to SLAVING\_IN\_PROGRESS and re-slaves until alignment is recovered before the operator can fire again.

### 12.2 Fire Command Sequence

- STEP 1 Operator presses Trigger Button (dedicated physical hardware button)
- STEP 2 Jetson FCC checks all fire entry conditions
- STEP 3 IF all conditions met:
  - Jetson FCC sends FIRE\_AUTHORIZATION (discrete signal) to ISM Safety MCU
  - ELSE:
    - Fire command rejected – display reason to operator
- STEP 4 ISM Safety MCU receives FIRE\_AUTHORIZATION

ISM independently verifies all armed conditions one final time (< 5 ms check)

STEP 5 IF ISM verification passes:

ISM pulses TRIGGER SOLENOID for the weapon-defined pulse duration

→ Trigger pin actuated → weapon fires

STEP 6 ISM monitors RECOIL MOVEMENT SENSOR:

IF recoil pulse detected within expected window:

→ Fire confirmed (mechanical verification)

→ ISM logs FIRE\_CONFIRMED event

ELSE:

→ Misfire detected → ISM sends MISFIRE\_ALERT to Jetson FCC

STEP 7 Jetson FCC transitions to STATE 7: FIRING → STATE 8: POST\_FIRE

### 12.3 Trigger Solenoid Pulse Parameters

| Parameter               | Value                                           |
|-------------------------|-------------------------------------------------|
| Pulse voltage           | 24 VDC                                          |
| Pulse duration          | Defined by weapon platform (typical: 50-200 ms) |
| Maximum duty cycle      | Single pulse per fire command                   |
| ISM verification window | Within 500 ms post-trigger                      |

## 12A. Mechanical Movement Verification Sensor

### 12A.1 Purpose & Concept

After every fire sequence, the weapon mechanism must complete a defined mechanical cycle — bolt carrier travel, ejection, and reload. A **dedicated short-travel displacement sensor** monitors a **20 mm mechanical movement** on the weapon action to verify that this full cycle has been completed correctly. This sensor is the gate that authorizes or blocks the next fire sequence.

This is a critical safety and reliability function. If the mechanical cycle did not complete — due to a misfire, jam, feed failure, or mechanical fault — the system **blocks all further fire commands** and alerts the operator immediately with a full fault description.

### 12A.2 Sensor Specification

| Parameter                    | Requirement                                                                     |
|------------------------------|---------------------------------------------------------------------------------|
| Sensor type                  | Linear displacement sensor (inductive proximity, Hall-effect magnetic, or LVDT) |
| Measurement range            | 0 - 25 mm (covers full 20 mm travel with margin)                                |
| Required travel confirmation | 20 mm $\pm$ 1 mm                                                                |
| Response time                | $\leq$ 5 ms detection latency                                                   |
| Output signal                | Discrete digital pulse (HIGH = travel complete) OR analog 0-5V                  |
| Interface                    | Discrete signal $\rightarrow$ ISM Safety MCU                                    |
| Operating environment        | IP54 minimum, vibration-rated                                                   |
| Mounting                     | Fixed to weapon frame, sensing element aligned to moving bolt/carrier component |
| Fail state                   | No signal = fault (active confirmation required, absence = block)               |

### 12A.3 Detection Logic — ISM Safety MCU

The ISM Safety MCU is the sole receiver and interpreter of the mechanical movement sensor signal. The logic is as follows:

AFTER TRIGGER SOLENOID PULSE:

START verification timer  $T_{\text{mech}}$  (timeout = 300 ms)

MONITOR mechanical movement sensor:

CASE 1 – MOVEMENT CONFIRMED:

- Sensor detects displacement  $\geq$  20 mm within  $T_{\text{mech}}$  window
  - $\rightarrow$  ISM sets flag: MECH\_CYCLE\_OK = TRUE
  - $\rightarrow$  ISM sends MECH\_CYCLE\_CONFIRMED to Jetson FCC
  - $\rightarrow$  GCS display: "CYCLE OK – READY FOR NEXT FIRE" in green
  - $\rightarrow$  Next fire sequence is AUTHORIZED

CASE 2 – MOVEMENT NOT DETECTED (Timeout):

- Sensor does NOT detect 20 mm displacement within  $T_{\text{mech}} = 300$  ms
  - $\rightarrow$  ISM sets flag: MECH\_CYCLE\_OK = FALSE
  - $\rightarrow$  ISM sets fault: MECH\_FAULT = JAM / INCOMPLETE\_CYCLE
  - $\rightarrow$  ISM sends MECH\_CYCLE\_FAULT to Jetson FCC
  - $\rightarrow$  ISM blocks all further ARMING\_AUTHORIZATION signals
  - $\rightarrow$  GCS display: FAULT BANNER – "WEAPON FAULT: MECHANICAL CYCLE INCOMPLETE"

→ Next fire sequence is BLOCKED until fault is cleared

CASE 3 – PARTIAL MOVEMENT (sensor detects < 20 mm):  
 Sensor detects movement but < 20 mm threshold  
 → Treated as CASE 2 (incomplete cycle)  
 → ISM sets fault: MECH\_FAULT = PARTIAL\_CYCLE  
 → Same blocking and alert behavior as CASE 2

#### 12A.4 Fault Classification & Operator Messages

| Fault Code   | Sensor Reading              | Displayed Message                                                           | Cause                         |
|--------------|-----------------------------|-----------------------------------------------------------------------------|-------------------------------|
| MECH-OK      | $\geq 20$ mm within 300 ms  | "CYCLE OK — READY FOR NEXT FIRE" (green)                                    | Normal operation              |
| MECH-JAM-01  | 0 mm — no movement detected | "WEAPON FAULT: NO MECHANICAL MOVEMENT — POSSIBLE JAM" (red flashing)        | Complete jam or feed failure  |
| MECH-JAM-02  | < 20 mm — partial movement  | "WEAPON FAULT: INCOMPLETE CYCLE — PARTIAL MOVEMENT DETECTED" (red flashing) | Partial jam or weak actuation |
| MECH-TIMEOUT | No signal within 300 ms     | "WEAPON FAULT: MECHANICAL VERIFICATION TIMEOUT" (red flashing)              | Sensor fault or disconnection |

#### 12A.5 Fault Clearance Procedure

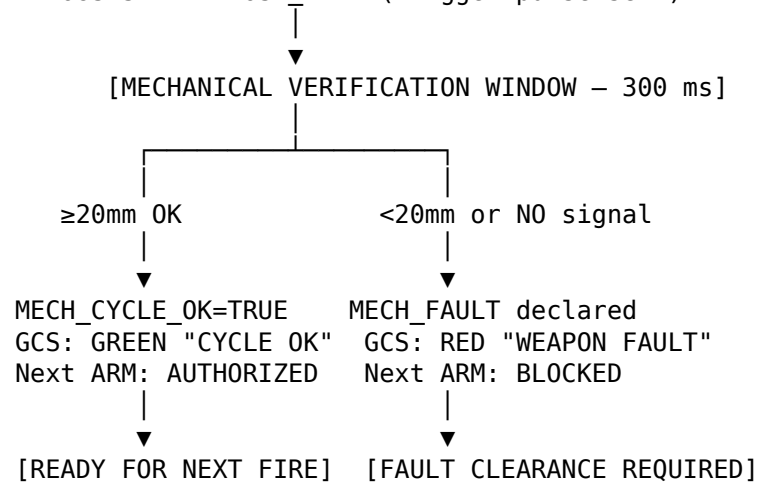
When a MECH\_FAULT is declared, the following conditions must ALL be met before the next fire sequence can be authorized:

STEP 1 Operator physically clears the weapon jam / confirms weapon state  
 STEP 2 Operator presses FAULT ACKNOWLEDGE on GCS (dedicated button or touchscreen)  
 STEP 3 ISM Safety MCU performs sensor re-check:  
     → Verifies mechanical movement sensor is reading zero (weapon in battery / closed)  
     → Verifies weapon temperature within limits  
     → Verifies recoil sensor nominal  
 STEP 4 IF all re-checks pass:  
     ISM clears MECH\_FAULT flag  
     ISM sends MECH\_FAULT\_CLEARED to Jetson FCC  
     System returns to BORESIGHT\_ACHIEVED state (camera still locked)  
     Operator may re-arm and fire  
 ELSE:  
     ISM remains in FAULT state  
     GCS continues to display fault banner  
     Operator must physically inspect weapon before next attempt

## 12A.6 Integration Into Fire Sequence — Entry Gate

The mechanical movement sensor status creates a new **mandatory gate** in the fire sequence:

PREVIOUS STATE: POST\_FIRE (trigger pulse sent)



## 12A.7 System Initialization Self-Test

During SYSTEM\_INIT (Phase 1), the ISM Safety MCU performs a self-test on the mechanical movement sensor:

STEP 1 ISM reads mechanical movement sensor at power-on  
 STEP 2 IF sensor reads 0 mm (weapon in closed/battery position): SENSOR\_OK → continue boot  
 STEP 3 IF sensor reads non-zero at rest: SENSOR\_FAULT → boot halted, alert displayed  
 STEP 4 Sensor is confirmed present and responding before IDLE\_SAFE state is entered

---

## **13. Phase 8 — Post-Fire, Mechanical Verification & Reset**

### **13.1 Immediate Post-Fire Actions (ISM Safety MCU)**

STEP 1 ISM de-energizes TRIGGER SOLENOID immediately after pulse  
STEP 2 ISM de-energizes SAFETY PIN SOLENOID → master trigger disarmed  
STEP 3 ISM energizes SAFETY CLOSE SOLENOID → closes weapon mechanical safety  
STEP 4 ISM transitions to SAFE state  
STEP 5 ISM STARTS mechanical verification window ( $T_{\text{mech}} = 300 \text{ ms}$ )  
→ Monitors mechanical movement sensor for  $\geq 20 \text{ mm}$  displacement  
STEP 6 ISM evaluates result (see Section 12A.3):  
IF MECH\_CYCLE\_OK = TRUE: MECH\_CYCLE\_CONFIRMED sent to Jetson FCC  
IF MECH\_CYCLE\_OK = FALSE: MECH\_FAULT declared, sent to Jetson FCC  
STEP 7 ISM sends SAFE\_CONFIRMED (discrete) to Jetson FCC

### **13.2 Post-Fire Actions (Jetson FCC)**

STEP 1 Jetson FCC receives SAFE\_CONFIRMED from ISM  
STEP 2 Jetson FCC logs fire event with timestamp, AZ, EL, target coordinates  
STEP 3 Jetson FCC reads WEAPON TEMPERATURE – if above  $T_{\text{warning}}$ : flag for operator  
STEP 4 Jetson FCC transitions to STATE 8: POST\_FIRE  
STEP 5 IMU STABILIZATION LOOP REMAINS ACTIVE during post-fire observation window  
→ Camera gimbal remains locked on target (for hit/miss confirmation per MIL-STD-1472H §5.13.1.5.1.4)  
→ Pan-tilt continues to be held on target coordinate by IMU corrections  
  
STEP 6 Jetson FCC receives MECH\_CYCLE result from ISM:  
  
CASE A – MECH\_CYCLE\_CONFIRMED (cycle OK):  
→ GCS displays "CYCLE OK – READY FOR NEXT FIRE" in green  
→ System confirms: next ARM sequence is authorized  
→ Operator may re-arm without clearance procedure  
→ State: BORESIGHT\_ACHIEVED maintained (camera still locked)  
  
CASE B – MECH\_FAULT received (cycle incomplete):  
→ GCS displays RED FAULT BANNER with fault code and message  
→ System BLOCKS next ARMING\_AUTHORIZATION  
→ IMU stabilization remains active (camera remains on target)  
→ Operator must acknowledge fault and follow clearance procedure (Section 12A.5)  
  
STEP 7 After operator acknowledges TARGET LOCK release (Button 1 again) or 5-second timeout:

\*\*\* IMU STABILIZATION LOOP DEACTIVATES \*\*\*  
 Gimbal Controller receives HOLD\_LAST\_POSITION  
 Worm gear passive self-lock holds pan-tilt at last position  
 Jetson FCC transitions to STATE 2: FREE\_OBSERVATION  
 → Camera lock is released → operator regains Joystick 1 control  
 → Pan-tilt may be commanded back to forward-horizon or remain at current position

### 13.3 Operator Display — POST\_FIRE

#### Case A — Mechanical cycle confirmed (normal):

|                                                                                   |                         |
|-----------------------------------------------------------------------------------|-------------------------|
| LIVE VIDEO FEED (E0 or IR)                                                        |                         |
|  | ← Amber bracket (5 sec) |
| FIRE COMPLETE                                                                     |                         |
| WEAPON: SAFE                                                                      | ✓ CYCLE OK              |
| READY FOR NEXT FIRE SEQUENCE                                                      |                         |
| IMU STABILIZATION: ACTIVE ●                                                       |                         |
| [OBSERVE FOR HIT/MISS – 5 SEC]                                                    |                         |
| [PRESS BTN1 TO RELEASE LOCK]                                                      |                         |

#### Case B — Mechanical fault (cycle incomplete):

|                                                                                     |                   |
|-------------------------------------------------------------------------------------|-------------------|
| LIVE VIDEO FEED (E0 or IR)                                                          |                   |
|  | ← Bracket remains |
| △ WEAPON FAULT – FIRE SEQUENCE BLOCKED △                                            |                   |
| [MECH-JAM-01: NO MECHANICAL MOVEMENT]                                               |                   |
| NEXT FIRE: BLOCKED                                                                  |                   |
| IMU STABILIZATION: ACTIVE ●                                                         |                   |
| CLEAR JAM → PRESS ACK → RE-ARM TO RESUME                                            |                   |

## 14. Operator Display Logic

### 14.1 HUD Symbol Summary

| Symbol                                           | Color              | Meaning                                                     | Phase              |
|--------------------------------------------------|--------------------|-------------------------------------------------------------|--------------------|
| Crosshair<br>(center)                            | White              | Camera<br>pointing<br>direction                             | All phases         |
| Target square                                    | White (static)     | Target locked<br>by camera                                  | Phase 3            |
| Route arrow                                      | White<br>animated  | Pan-tilt slaving<br>in progress                             | Phase 4            |
| $\Delta$ AZ / $\Delta$ EL<br>readout             | White              | Live angular<br>error values                                | Phase 4            |
| "IMU STABI-<br>LIZATION:<br>STANDBY"             | Grey               | IMU not yet<br>active                                       | Phase 4            |
| Target square                                    | Red (blinking)     | Boresight<br>achieved —<br>aligned                          | Phase 5<br>onwards |
| "IMU STABI-<br>LIZATION:<br>ACTIVE ●"            | Green pulsing      | IMU actively<br>correcting<br>AZ/EL servos                  | Phase 5, 6, 7, 8   |
| "WEAPON<br>ARMED"                                | Red                | System armed,<br>ready to fire                              | Phase 6            |
| banner<br>"CYCLE OK —<br>READY FOR<br>NEXT FIRE" | Green              | Mechanical<br>cycle<br>confirmed                            | Phase 8<br>normal  |
| "WEAPON<br>FAULT: [code]"                        | Red flashing       | Mechanical<br>cycle<br>incomplete —<br>next fire<br>blocked | Phase 8 fault      |
| "FIRE<br>COMPLETE"                               | Amber              | Post-fire<br>confirmation                                   | Phase 8            |
| banner<br>"IMU STABI-<br>LIZATION:<br>OFF"       | Grey               | Target lock<br>released, IMU<br>loop closed                 | FREE_OBSERVATION   |
| Fault banner                                     | Flashing<br>orange | Any fault or<br>abort<br>condition                          | Phase F            |

## 14.2 Pan-Tilt Route Display Logic

During SLAVING\_IN\_PROGRESS:

- The GCS overlays the current pan-tilt position as a secondary

- crosshair or dot on the compass/angular display
- An animated line or arrow connects the pan-tilt current position to the target locked square
- As  $\Delta AZ$  and  $\Delta EL$  decrease, the animated indicator visually closes the gap toward the target
- Numerical  $\Delta AZ$  and  $\Delta EL$  update in real time (minimum 10 Hz refresh)

### 14.3 Alignment Confirmation Indication

When boresight is achieved:

- The target square changes to **blinking red at 2 Hz**
- The  $\Delta AZ$  and  $\Delta EL$  readouts change to green and show values  $\leq$  threshold
- An audio tone (if implemented) sounds once
- The "WEAPON ARMING NOW AVAILABLE" label appears

## 15. Abort & Safety Cutoff Logic

### 15.1 Abort Triggers

The system **immediately transitions to FAULT/ABORT state** on any of the following events, regardless of current state:

| Event                                                | Source                | System Response                           |
|------------------------------------------------------|-----------------------|-------------------------------------------|
| RF link loss > 2 seconds                             | Jetson FCC RF monitor | Weapon SAFE, servos hold position         |
| Jetson A watchdog timeout                            | Hardware watchdog     | Failover to Jetson B; weapon SAFE         |
| Weapon temperature > $T_{cutoff}$                    | ISM Safety MCU        | ISM forces SAFE, fire rejected            |
| Recoil sensor anomaly (unexpected fire)              | ISM Safety MCU        | ISM forces SAFE, MISFIRE alert            |
| Mechanical movement sensor fault (no signal at boot) | ISM Safety MCU        | Boot halted, sensor fault alert           |
| MECH_FAULT — incomplete cycle post-fire              | ISM Safety MCU        | Next fire BLOCKED, fault banner displayed |
| AZ/EL sensor fault                                   | Jetson FCC            | Servos halt, weapon SAFE                  |

| Event                                                        | Source     | System Response                  |
|--------------------------------------------------------------|------------|----------------------------------|
| ISM heartbeat loss                                           | Jetson FCC | Weapon SAFE,<br>operator alert   |
| Operator presses<br>ABORT button                             | GCS        | Immediate SAFE<br>state          |
| Alignment lost after<br>BORESIGHT ( $ \Delta  > 0.5^\circ$ ) | Jetson FCC | Exit ARMED,<br>re-slave required |

## 15.2 Abort Sequence

- STEP 1 Abort trigger detected
- STEP 2 Jetson FCC sends ABORT command to ISM (discrete signal)
- STEP 3 ISM immediately:
  - De-energizes all solenoids
  - Energizes SAFETY CLOSE solenoid
  - Transitions to SAFE state
- STEP 4 Gimbal Controller: AZ and EL servos commanded to hold position (no runaway)
- STEP 5 GCS display: Flashing orange FAULT banner with fault reason text
- STEP 6 Camera: Remains locked on last position (if camera lock was active)
- STEP 7 System remains in FAULT state until:
  - Fault condition cleared
  - Operator manually acknowledges fault on GCS
  - System transitions to IDLE\_SAFE

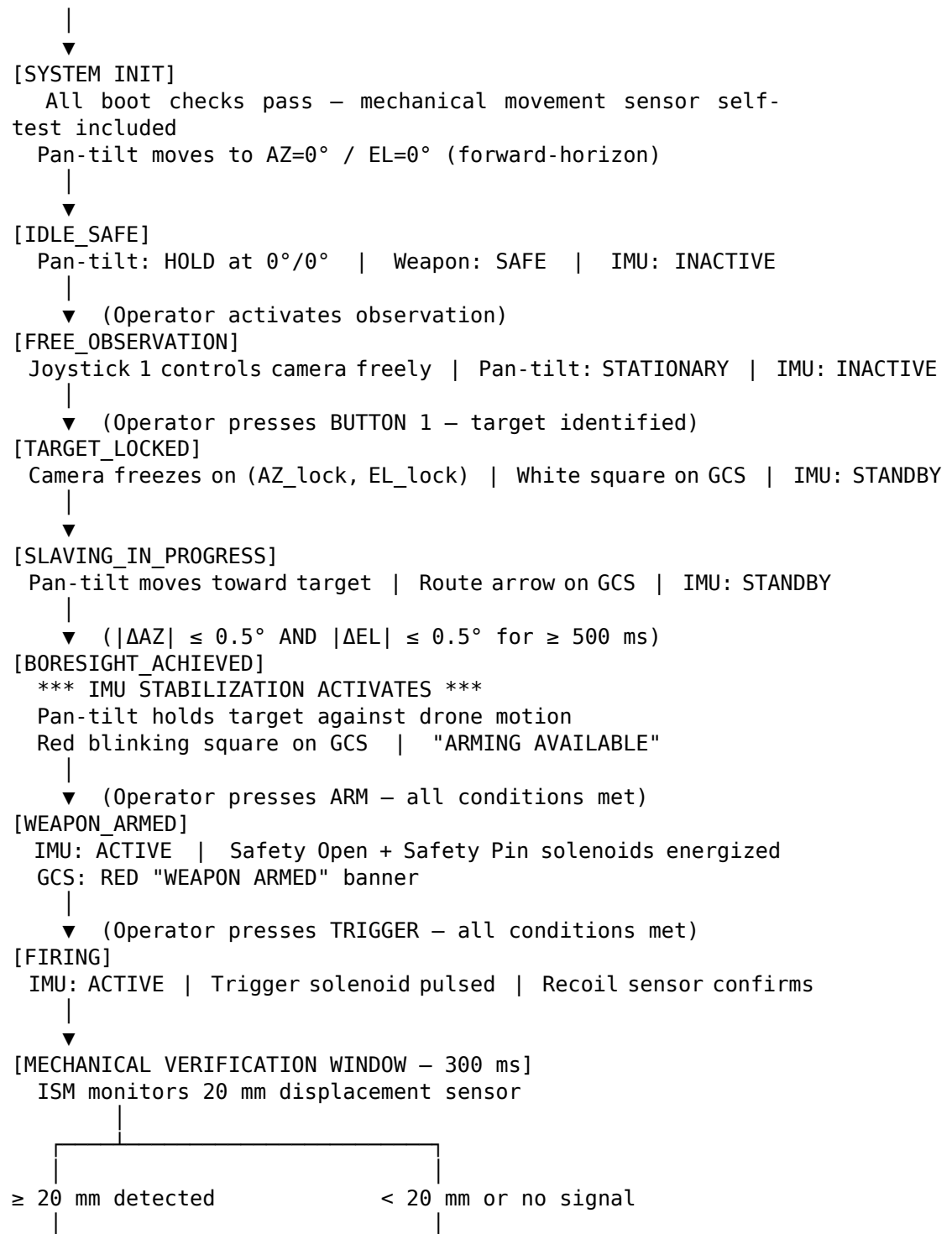
## 15.3 Non-Overridable Safety Rules

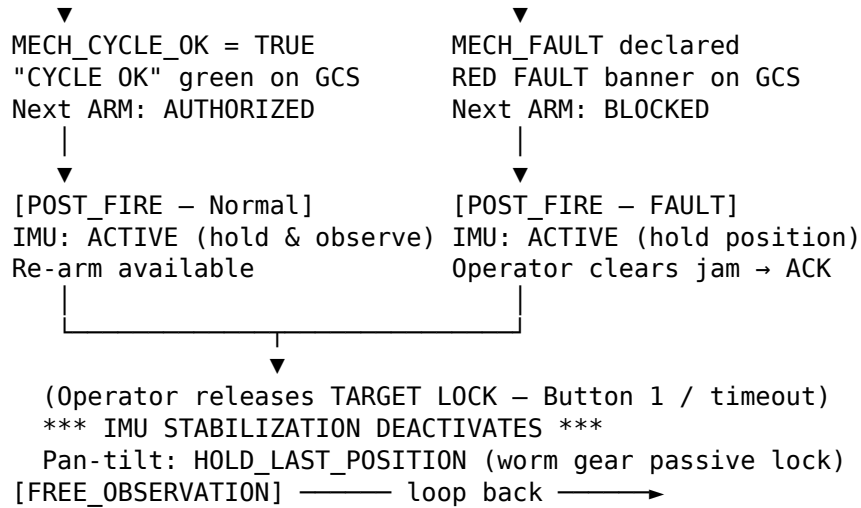
The following rules are **hardcoded in the ISM Safety MCU and cannot be overridden by software**:

1. The trigger solenoid can **never** be energized unless the ISM is in ARMED state
2. The ARMED state can **never** be entered unless the BORE-SIGHT\_ACHIEVED signal is present
3. Any loss of the ISM's own heartbeat causes all solenoids to de-energize (fail-safe design)
4. De-energized solenoid = SAFE mechanical state (spring-return to safe)

## 16. Full Sequence Flow Diagram (Text Representation)

POWER ON





**Two governing rules — always apply:** 1. TARGET\_LOCK engaged → IMU stabilization ACTIVE. TARGET\_LOCK released → INACTIVE. 2. MECH\_CYCLE\_OK = TRUE → next fire AUTHORIZED. MECH\_CYCLE\_OK = FALSE → next fire BLOCKED.

## 17. Interface & Signal Summary

| Signal Name           | Direction                              | Type                 | Description                                           |
|-----------------------|----------------------------------------|----------------------|-------------------------------------------------------|
| AZ_cam,<br>EL_cam     | Camera →<br>Jetson FCC                 | Ethernet (100<br>Hz) | Camera<br>current<br>pointing<br>angles               |
| TARGET_LOCK_CONFIRMED | Camera →<br>Jetson FCC                 | Ethernet             | Lock<br>coordinates<br>confirmed                      |
| IMU angular<br>rates  | IMU → Jetson<br>FCC                    | SPI/CAN (1<br>kHz)   | $\omega_x, \omega_y, \omega_z$<br>drone body<br>rates |
| AZ_pantilt            | AZ Sensor →<br>Jetson FCC              | CAN (100 Hz)         | Pan-tilt<br>azimuth<br>position                       |
| EL_pantilt            | EL Sensor →<br>Jetson FCC              | CAN (100 Hz)         | Pan-tilt<br>elevation angle                           |
| EL_direction          | EL Direction<br>Sensor →<br>Jetson FCC | CAN                  | Elevation<br>direction                                |

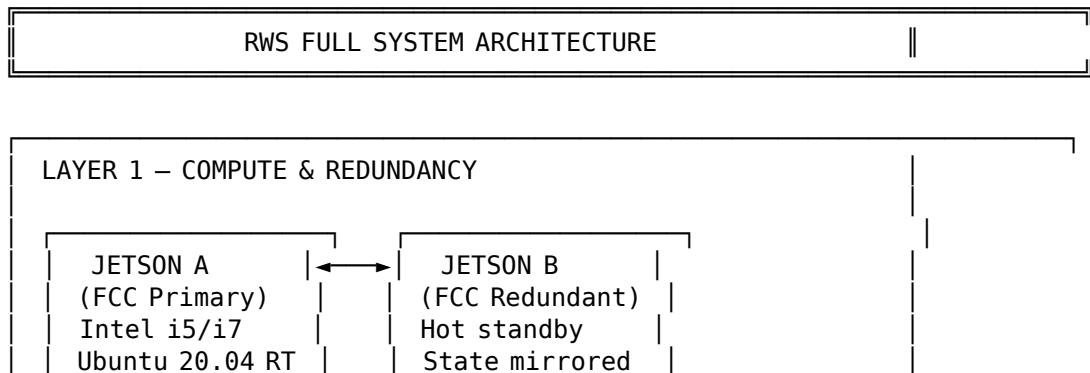
| Signal Name             | Direction                      | Type               | Description                                   |
|-------------------------|--------------------------------|--------------------|-----------------------------------------------|
| AZ_command              | Jetson FCC → Gimbal Controller | CAN                | Target azimuth for slaving / IMU correction   |
| EL_command              | Jetson FCC → Gimbal Controller | CAN                | Target elevation for slaving / IMU correction |
| AZ_motor_cmd            | Gimbal Controller → AZ Servo   | CAN (Motor ctrl)   | AZ servo velocity command                     |
| EL_motor_cmd            | Gimbal Controller → EL Servo   | CAN (Motor ctrl)   | EL servo velocity command                     |
| BORESIGHT_ALIGNMENT     | Jetson FCC → ISM               | Discrete (24V)     | Alignment confirmed — IMU active              |
| ARMING_AUTHORIZATION    | Jetson FCC → ISM               | Discrete (24V)     | Arm command from FCC                          |
| ARMED_CONFIRMATION      | ISM → Jetson FCC               | Discrete (24V)     | ISM confirms armed state                      |
| FIRE_AUTHORIZATION      | Jetson FCC → ISM               | Discrete (24V)     | Fire command from FCC                         |
| Safety Open cmd         | ISM → Safety Open Solenoid     | Discrete 24V       | Energize = open safety                        |
| Safety Close cmd        | ISM → Safety Close Solenoid    | Discrete 24V       | Energize = close safety                       |
| Safety Pin cmd          | ISM → Safety Pin Solenoid      | Discrete 24V       | Energize = arm master trigger                 |
| Trigger cmd             | ISM → Trigger Solenoid         | Discrete 24V pulse | Energize = fire weapon                        |
| Recoil pulse            | Recoil Sensor → ISM            | Discrete           | Fire actuation mechanical verification        |
| Mech movement signal    | Movement Sensor → ISM          | Discrete / Analog  | 20 mm cycle verification — HIGH = complete    |
| MECH_CYCLE_CONFIRMATION | ISM → Jetson FCC               | Discrete (24V)     | Cycle OK — next fire authorized               |

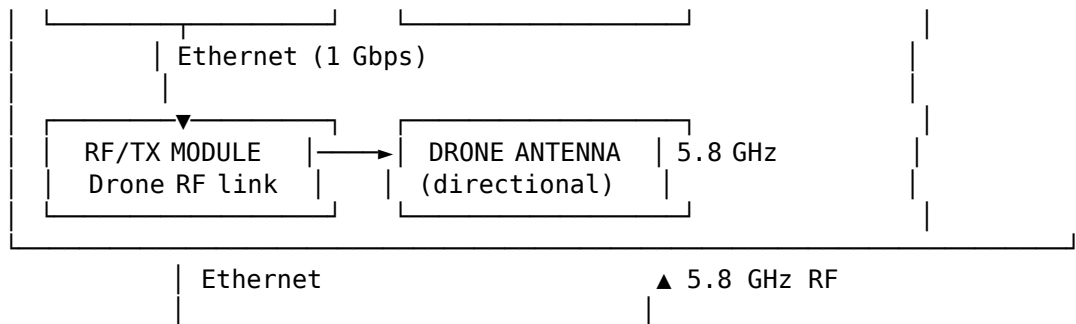
| Signal Name              | Direction                             | Type            | Description                                |
|--------------------------|---------------------------------------|-----------------|--------------------------------------------|
| MECH_FAULT               | ISM → Jetson<br>FCC                   | Discrete (24V)  | Cycle incomplete —<br>next fire<br>blocked |
| Weapon temp              | Temp Sensor →<br>ISM                  | Analog/CAN      | Weapon barrel<br>temperature               |
| Laser range              | Laser Range<br>Finder →<br>Jetson FCC | Serial/Ethernet | Target<br>distance in<br>meters            |
| Joystick 1               | GCS → Jetson<br>FCC                   | RF / Ethernet   | Camera slew<br>command                     |
| Button 1                 | GCS → Jetson<br>FCC                   | RF / Ethernet   | Target lock<br>activation /<br>release     |
| Arm Button<br>(BTN2)     | GCS → Jetson<br>FCC                   | RF / Ethernet   | Weapon arm<br>command                      |
| Trigger Button<br>(BTN6) | GCS → Jetson<br>FCC                   | RF / Ethernet   | Fire command                               |
| Display data             | Jetson FCC →<br>GCS                   | Ethernet / RF   | All HUD<br>sybology and<br>video           |

## 18. Full System Hardware Architecture

### 18.1 Architecture Overview

The following is the complete hardware node diagram of the RWS system, showing every physical component, its role, and all interconnections.





## LAYER 2 – SAFETY LAYER

ISM SAFETY MCU (Independent – NOT on compute bus)  
STM32H7 or TI TMS570 – ISO 13849-1 PL-d

### INPUTS (discrete):

- BORESIGHT\_ACHIEVED ◀ Jetson A
- ARMING\_AUTHORIZATION ◀ Jetson A
- FIRE\_AUTHORIZATION ◀ Jetson A
- Recoil Sensor →
- Mech Movement Sensor (20mm) →
- Weapon Temp Sensor →

### OUTPUTS (discrete 24V):

- Safety Open Solenoid
- Safety Close Solenoid
- Safety Pin Solenoid
- Trigger Solenoid

### Outputs to Jetson A (discrete):

- ARMED\_CONFIRMED, MECH\_CYCLE\_CONFIRMED, MECH\_FAULT, SAFE\_CONFIRMED

## LAYER 3 – OPTICS & ELECTRO-OPTICAL

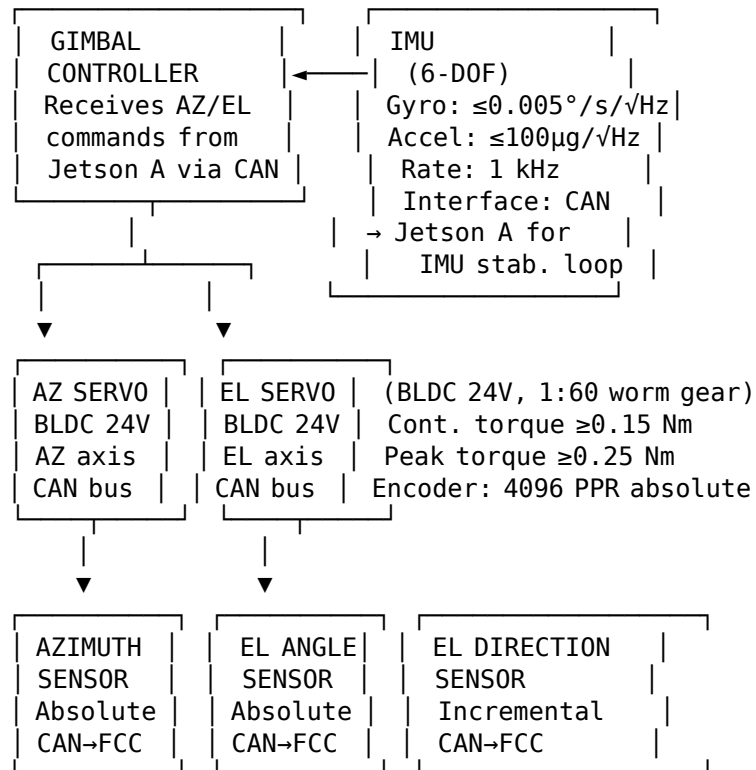
OPTIC MASTER  
(Camera PC)  
Connected to  
Jetson A via  
Ethernet  
GigE, H.264/265

EO DAY CAMERA  
(Day Payload) Full HD 1080p  
10x optical zoom  
IR THERMAL CAM  
(IR Payload) 640x512 LWIR  
NETD ≤ 50 mK

EO/IR CAMERA GIMBAL (mounted on RWS Elevation Platform)  
– Self-contained stabilized gimbal (own motors + own IMU)

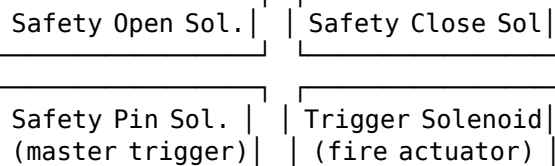
- Master in the master/slave fire control loop
- Reports AZ\_cam, EL\_cam to Jetson A via Ethernet at 100 Hz
- Receives Joystick 1 slew commands from GCS via Jetson A

#### LAYER 4 – MOTION CONTROL & SENSOR FEEDBACK



#### LAYER 5 – WEAPON MECHANICS & SENSORS

SOLENOIDS (all 24V, controlled by ISM Safety MCU only):



SENSORS (all report to ISM Safety MCU):

RECOIL MOVEMENT SENSOR

– Discrete pulse → ISM – confirms weapon fired mechanically

MECHANICAL MOVEMENT SENSOR (20 mm displacement sensor)

– Inductive / Hall-effect / LVDT  
 – Measures bolt/carrier 20 mm travel post-fire  
 – Discrete HIGH ( $\geq 20\text{mm}$ ) = cycle complete → next fire OK  
 – No signal or  $< 20\text{mm}$  = cycle fault → next fire BLOCKED  
 – Self-test at SYSTEM\_INIT before IDLE\_SAFE

WEAPON TEMPERATURE SENSOR

– Analog / digital temperature → ISM  
 – Blocks fire if temp  $> T_{\text{cutoff}}$

LASER RANGE FINDER

– Serial / Ethernet → Jetson A  
 – Provides target distance for ballistic correction

LAYER 6 – GROUND CONTROL STATION (GCS)

Mobilicom 10" Maxi Controller Pro

Panasonic Toughpad FZ-G2 – 1920×1200 – Intel Core i5/i7

Windows 10 Pro – Integrated Mobilicom RF/TX

CONTROLS:

Joystick 1 → Camera Gimbal AZ/EL slew

Joystick 2 → Reserved (manual override / maintenance)

BTN 1 → Target Lock / Unlock

BTN 2 (SG) → Arm Weapon (safeguard)

BTN 3 → Emergency Abort

BTN 4 → Camera mode (E0/IR/PIP)

BTN 5 → Zoom control

BTN 6 (SG) → Fire Trigger (safeguard)

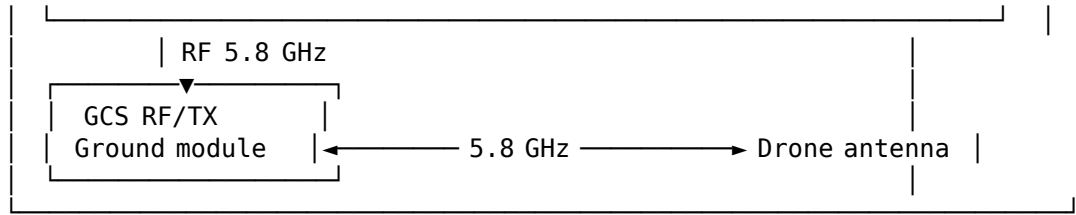
SLI 1 → E0/IR feed toggle

SLI 2 → Zoom level

SLI 3 → IR gain/brightness

TOG 1 → System master power

TOG 2 (SG) → Weapon master enable (physical dual interlock)



## 18.2 Bus Architecture Summary

| Bus             | Physical                                | Devices                                                                                             | Data Rate              |
|-----------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------|
| Ethernet (GigE) | CAT6 shielded                           | Jetson A ↔<br>Jetson B ↔<br>Optic Master<br>↔ Laser Range<br>Finder                                 | 1 Gbps                 |
| CAN-FD          | Shielded<br>twisted pair,<br>120Ω term. | Jetson A ↔<br>Gimbal<br>Controller ↔<br>AZ Servo ↔ EL<br>Servo ↔ AZ<br>Sensor ↔ EL<br>Sensors ↔ IMU | 5 Mbps                 |
| Discrete 24V    | Individual<br>signal wires              | Jetson A ↔ ISM<br>↔ All Solenoids<br>↔ Recoil<br>Sensor ↔<br>Mech<br>Movement<br>Sensor             | N/A                    |
| RF 5.8 GHz      | Antenna<br>(2×SDR)                      | Drone antenna<br>↔ GCS RF/TX                                                                        | ≥10 Mbps               |
| Video           | H.264/H.265<br>over<br>Ethernet/RF      | Optic Master<br>→ Jetson A →<br>GCS                                                                 | ≥8 Mbps per<br>channel |
| SPI             | PCB trace                               | IMU → Jetson<br>A (high-rate<br>IMU data<br>path)                                                   | 10 Mbps                |

## 18.3 Power Distribution

| Rail          | Source                | Consumers             | Protection                   |
|---------------|-----------------------|-----------------------|------------------------------|
| 24V Main Bus  | External or battery   | All subsystems        | Main fuse + reverse polarity |
| 12V Compute   | 24V → DC-DC isolated  | Jetson A, Jetson B    | Isolated filtered rail       |
| 24V Servo     | 24V direct            | AZ servo, EL servo    | Per-servo current limit      |
| 24V Solenoid  | 24V direct            | All 4 solenoids       | ISM-switched, fused          |
| 5V Safety     | 24V → LDO independent | ISM Safety MCU only   | Independent regulator        |
| 12V/5V Optics | 24V → DC-DC           | Optic Master, cameras | Low-noise LDO                |

#### 18.4 Complete Component List

| # | Component                | Qty | Interface               | Location            |
|---|--------------------------|-----|-------------------------|---------------------|
| 1 | Jetson A (FCC Primary)   | 1   | Ethernet, CAN, Discrete | RWS body            |
| 2 | Jetson B (FCC Redundant) | 1   | Ethernet                | RWS body            |
| 3 | RF/TX Drone Module       | 1   | RF 5.8 GHz              | RWS body            |
| 4 | Drone Antenna            | 1   | RF 5.8 GHz              | RWS top             |
| 5 | ISM Safety MCU           | 1   | Discrete 24V            | RWS body (isolated) |
| 6 | Optic Master (Camera PC) | 1   | Ethernet, MIPI/GigE     | EL platform         |
| 7 | EO Day Camera            | 1   | GigE / MIPI             | EL platform         |
| 8 | IR Thermal Camera        | 1   | GigE / MIPI             | EL platform         |
| 9 | EO/IR Camera Gimbal      | 1   | Ethernet (coords)       | EL platform         |

| #  | Component                                    | Qty | Interface            | Location         |
|----|----------------------------------------------|-----|----------------------|------------------|
| 10 | IMU<br>(6-DOF)                               | 1   | CAN / SPI            | RWS body         |
| 11 | Gimbal<br>Controller                         | 1   | CAN                  | RWS body         |
| 12 | AZ Servo<br>(BLDC<br>24V)                    | 1   | CAN                  | AZ axis          |
| 13 | EL Servo<br>(BLDC 24V,<br>1:60 worm)         | 1   | CAN                  | EL axis          |
| 14 | Azimuth<br>Sensor<br>(absolute)              | 1   | CAN                  | AZ axis          |
| 15 | EL Angle<br>Sensor<br>(absolute)             | 1   | CAN                  | EL axis          |
| 16 | EL<br>Direction<br>Sensor (in-<br>cremental) | 1   | CAN                  | EL axis          |
| 17 | Safety<br>Open<br>Solenoid                   | 1   | Discrete<br>24V      | Weapon<br>mount  |
| 18 | Safety<br>Close<br>Solenoid                  | 1   | Discrete<br>24V      | Weapon<br>mount  |
| 19 | Safety Pin<br>Solenoid                       | 1   | Discrete<br>24V      | Weapon<br>mount  |
| 20 | Trigger<br>Solenoid                          | 1   | Discrete<br>24V      | Weapon<br>mount  |
| 21 | Recoil<br>Movement<br>Sensor                 | 1   | Discrete             | Weapon<br>mount  |
| 22 | Mechanical<br>Movement<br>Sensor<br>(20mm)   | 1   | Discrete /<br>Analog | Weapon<br>action |
| 23 | Weapon<br>Tempera-<br>ture Sensor            | 1   | Analog /<br>CAN      | Weapon<br>barrel |
| 24 | Laser<br>Range<br>Finder                     | 1   | Serial /<br>Ethernet | EL<br>platform   |

| #  | Component                             | Qty | Interface  | Location            |
|----|---------------------------------------|-----|------------|---------------------|
| 25 | GCS —<br>Mobilicom<br>10" Maxi<br>Pro | 1   | RF 5.8 GHz | Operator<br>station |
| 26 | GCS RF/TX<br>Ground<br>Module         | 1   | RF 5.8 GHz | Operator<br>station |

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*End of Document — RWS-SW-LOGIC-002 Rev B*

*This document must be reviewed by the system safety engineer before any implementation begins. All state transition thresholds (0.5° alignment, 500 ms hold, 20 mm travel, 300 ms verification window, T\_cutoff values) are subject to revision during system integration testing.*